

FINAL
CAPSTONE

Transaction processing, made inspectable

Validation · Fraud scoring ·
Settlement

Vladyslav Shmygelskyy

August 2026

Three stages turn raw records into explicit decisions

01 Validate

Required fields, Decimal amounts, supported ISO 4217 codes, and timezone-aware timestamps.

02 Score risk

High value, unusual UTC hours, and cross-border USD indicators produce a transparent 0–100 score.

03 Settle

Rejected records stop; high risk is held; all other validated records settle deterministically.

INPUT

DECISION

RESULT

Every sample transaction reaches a final outcome

Verified pipeline run

All eight inputs produced privacy-safe JSON results and a summary consumed by the dashboard and MCP server.

5

settled

1

held for review

2

rejected

The file protocol makes each transition observable

Standard envelopes

UUID message ID, UTC timestamp, source, target, type, and transaction data accompany each transition.

Privacy boundary

Final results omit descriptions and replace plaintext accounts with derived references. Audit logs use allowlisted fields.

Two query surfaces

A FastAPI dashboard serves people; FastMCP tools and `pipeline://summary` serve AI clients.

●
TRACE

●
PROTECT

●
OBSERVE

Automation turns quality goals into a release gate

Evidence, not assertion

Unit and integration tests run in isolated temporary directories. The Git pre-push hook invokes the same enforced pytest coverage configuration.

27

tests passed

97.84%

statement coverage

80%

minimum push gate

Clear contracts make agentic work verifiable

Specify decisions before
generating code.
Keep domain logic separate
from interfaces.

**Automate the evidence
reviewers need.**